AMD Unsloth Studio Fine-tuning

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Notebook access

notebooks.amd.com/ind-dev

Password: RadeonUnslothTuning

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Agenda

01 **Fine-tuning: what and when**
The problem it solves

02 **Types of fine-tuning**
LoRA, QLoRA, SFT, GRPO

03 **The Unsloth advantage**
Faster, leaner, no-code

04 **Support tickets to JSON**
Our supervised task

05 **AMD Radeon GPU**
rocm-smi and rocminfo

06 **Hands-on in Studio**
Train, compare, score, export

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Part 1: The Concepts

What a model is, what fine-tuning changes, and how SFT learns from examples

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What is a Large Language Model?

A neural network trained on a huge amount of text so it can predict **the next token** given what came before.

This ability enables models to chat, summarize, write code and reason.



Base models

Raw next-token predictors. Great starting point for training but awkward to chat with.

Instruct models

Further trained to follow instructions and hold a conversation.

Generalists

Llama, Qwen, Mistral on Hugging Face know a bit of everything, but not something specific to you.

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What is fine-tuning? Why do it?

Take a pre-trained model and keep training it a little more on **YOUR** data, so it adapts to **YOUR** task, domain, tone or format.

1

Voice

Speak in a specific style or persona: your brand, your support agent.

2

Domain

Know legal, medical, or your own product documentation.

3

Format

Always return JSON. Always cite sources.

4

Focus

Do one narrow task extremely well, cheaply, on a small model.

Often cheaper and more reliable than stuffing everything into a giant prompt or paying for an even bigger model. **It lets a small model punch above its weight.**

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LoRA and QLoRA

Parameter-efficient fine-tuning: freeze the giant model, train a tiny adapter.

LoRA Low-Rank Adaptation

- Inject small low-rank matrices into chosen layers
- Original weights stay frozen, only adapters are trained
- Trainable parameters drop by orders of magnitude

QLoRA Quantized Low-Rank Adaptation

- Everything LoRA does, plus a 4-bit base model
- Cuts VRAM again, so bigger models fit on one GPU
- The usual answer to an out-of-memory error

~0.1%

of parameters trained

4-bit

base weights under QLoRA

Comparable

quality to full fine-tuning

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Training methodology: SFT and GRPO

Two ways to tell a model it did well. SFT shows the answer, while GRPO scores attempts.

SFT Supervised Fine-Tuning

- Uses labelled data: prompt in, correct answer out
- The model imitates the answers you supply

```
{"instruction": "Summarize in one line.",
  "input":  "<article text>",
  "output": "<summary>"}
```

GRPO Group Relative Policy Optimization

- No labelled answer needed, only a way to score one
- Model generates a group of attempts and compares them

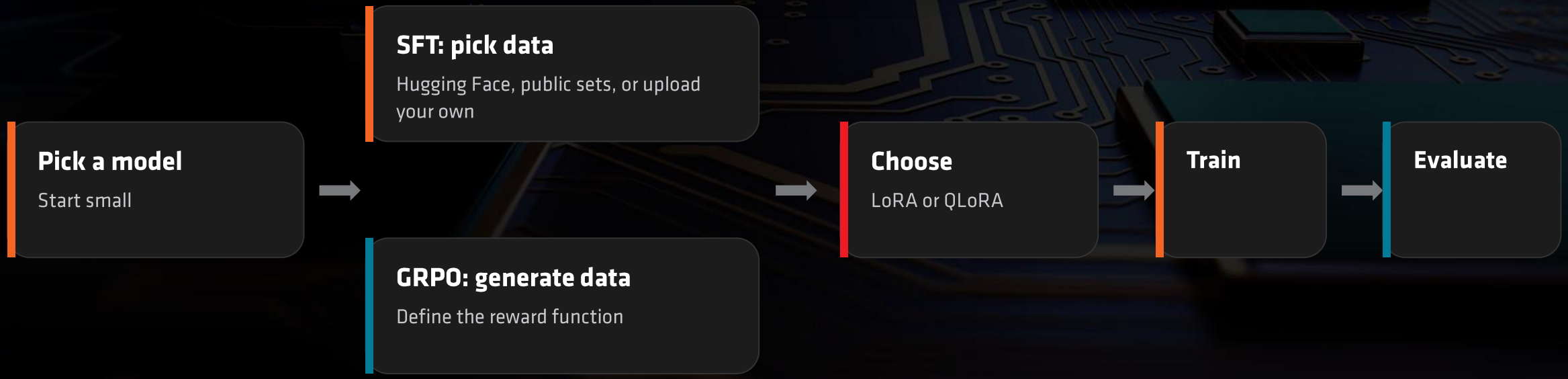
1. **Generate** several candidate answers
2. **Score** each with a reward function
3. **Prefer** the ones that scored above the group

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The fine-tuning pipeline

Same spine either way. The data step is what differs.



Training knobs

`learning rate` `number of steps` `time limit` `max sequence length`

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Unsloth

An open-source, no-code framework and UI for training, running and exporting LLMs **locally**, tuned for speed, memory efficiency and accessibility.

It sits as an optimized layer on top of Hugging Face Transformers, so your existing workflow still applies.

LoRA and QLoRA

built-in

Quantization

4-bit and 8-bit paths

Optimized kernels

faster attention and MLP

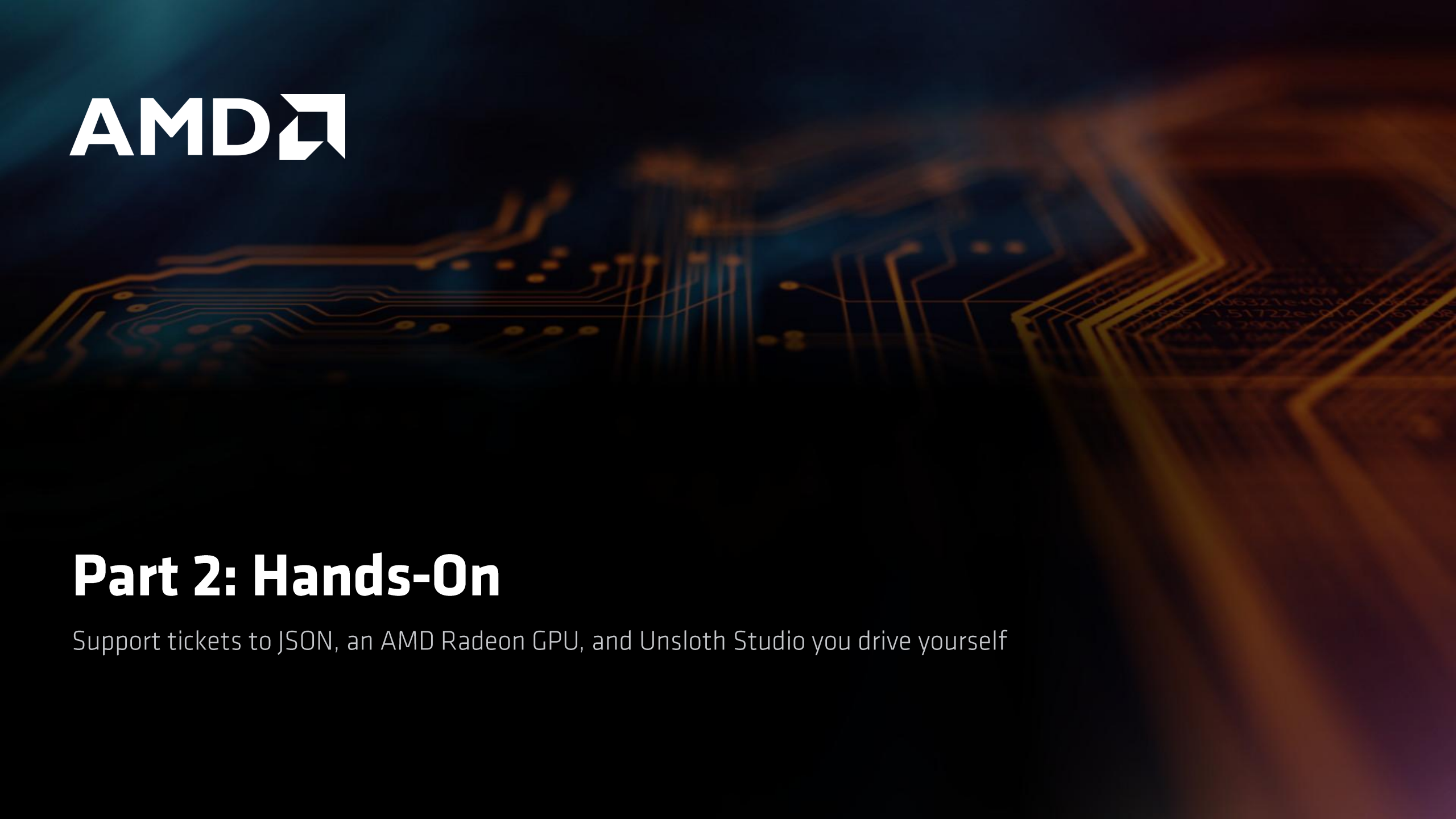
Consumer hardware

one GPU is enough

In this workshop

Unsloth Studio puts the whole loop in a browser tab: pick a model, upload data, train, chat, export. No code.

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Part 2: Hands-On

Support tickets to JSON, an AMD Radeon GPU, and Unsloth Studio you drive yourself

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Support tickets to JSON

Messy text in, **structured data** out.

Customers write however they like. The ticket system needs the same seven fields, in the same format, every time.

Verifiable

Every field is right or wrong. Score it with code.

Teaches format

One-line prompt in, exact schema out. No rules in the prompt.

Real judgment

“Not the model I ordered” means wrong_item. Dates and amounts get normalized.

SUPPORT MESSAGE

Case #99570. Customer Elena reports the rice cooker is not the model I ordered (ref ORD-10968). Order placed 24 January 2026. Total was USD 867.25. Call me on (815) 781-9664.

fine-tuned Qwen3.5

TICKET JSON

```
{
  "order_id": "ORD-10968",
  "customer": "Elena",
  "phone": "(815) 781-9664",
  "product": "rice cooker",
  "category": "wrong_item",
  "purchase_date": "2026-01-24",
  "amount": 867.25
}
```

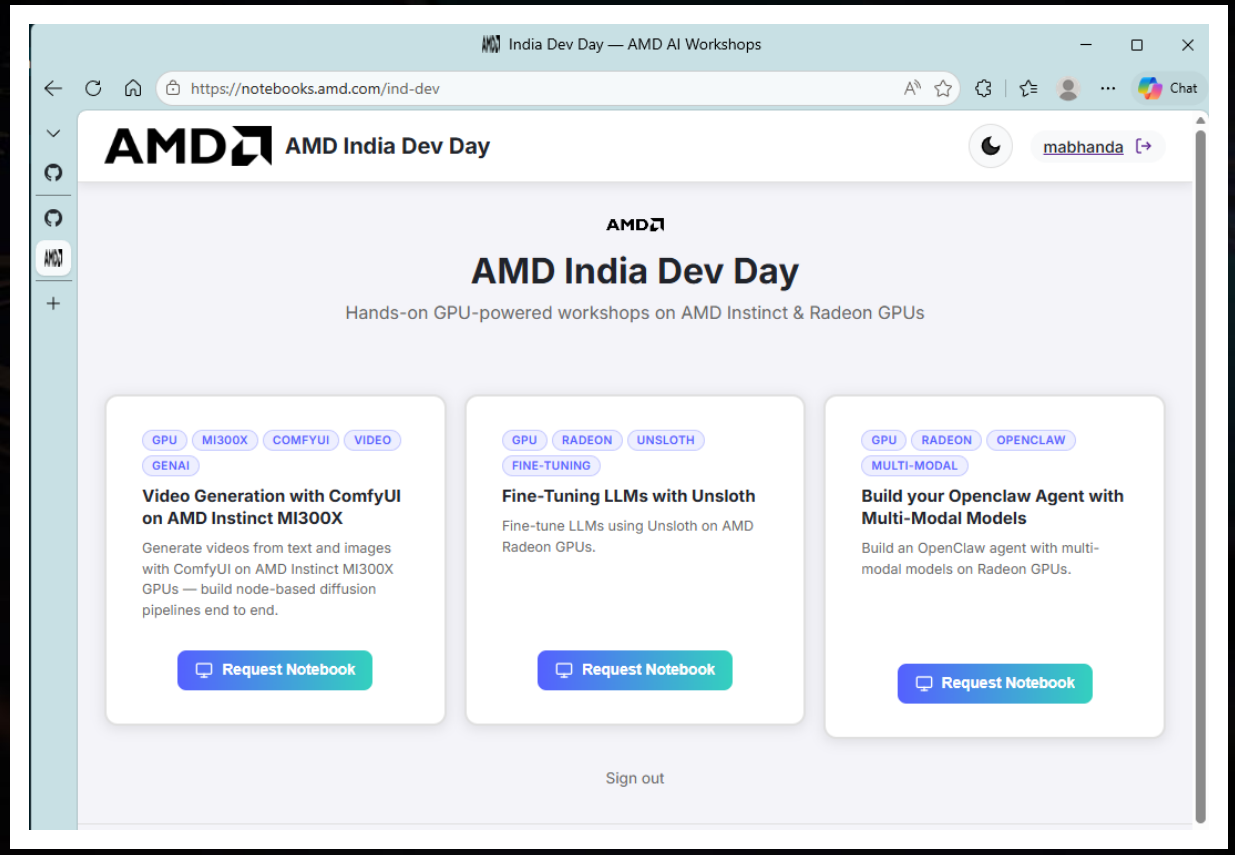
A held-out message and the fine-tuned model’s answer

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AMD Unsloth fine-tuning workshop

1. **Open the portal**
   `notebooks.amd.com/ind-dev`
2. **Enter the password**
3. **Pick the Unsloth card**
   Lands you on an AMD Radeon GPU



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GPU visibility and health

Open a terminal inside Jupyter and confirm the GPU is available before you train.

terminal

```
$ rocm-smi
$ rocminfo
```

rocm-smi

Live view: temperature, power, clocks, VRAM use

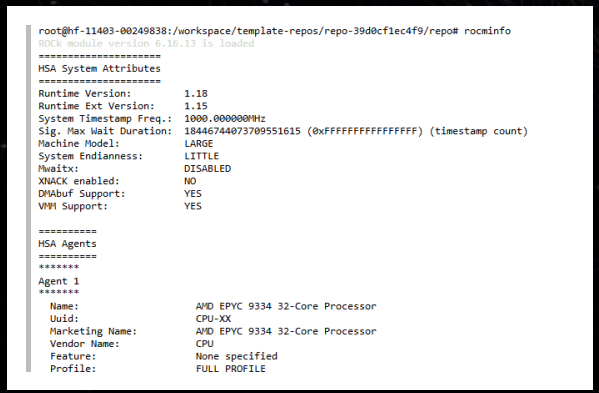
rocminfo

Static view: agents, ROCm version, GPU identity

GPU busy at 0% while a cell runs? It is not training.

```
root@hf-11403-00249838:/workspace/template-repos/repo-39d0cf1ec4f9/repo# rocm-smi


========================================= ROCm System Management Interface =========================================
=
================================================== Concise Info ==================================================
=
Device  Node  IDs              Temp    Power  Partitions          SCLK  MCLK   Fan    Perf  PwrCap  VRAM%  GPU%
              (DID,     GUID)  (Edge)  (Avg)  (Mem, Compute, ID)
==================================================================================================================
=
0       3     0x744b,   49884  27.0°C  7.0W   N/A, N/A, 0         0Mhz  96Mhz  20.0%  auto  241.0W  0%     0%
==================================================================================================================
=
============================================== End of ROCm SMI Log ===============================================
=
root@hf-11403-00249838:/workspace/template-repos/repo-39d0cf1ec4f9/repo#
```



rocminfo confirms the ROCm runtime version and every agent the host exposes, GPU and CPU alike.

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Driving Studio

Four steps on the Train page. The first model load is the slow one.

Pick model and method

`unsloth/Qwen3.5-2B`, method `LoRA`

Upload the data

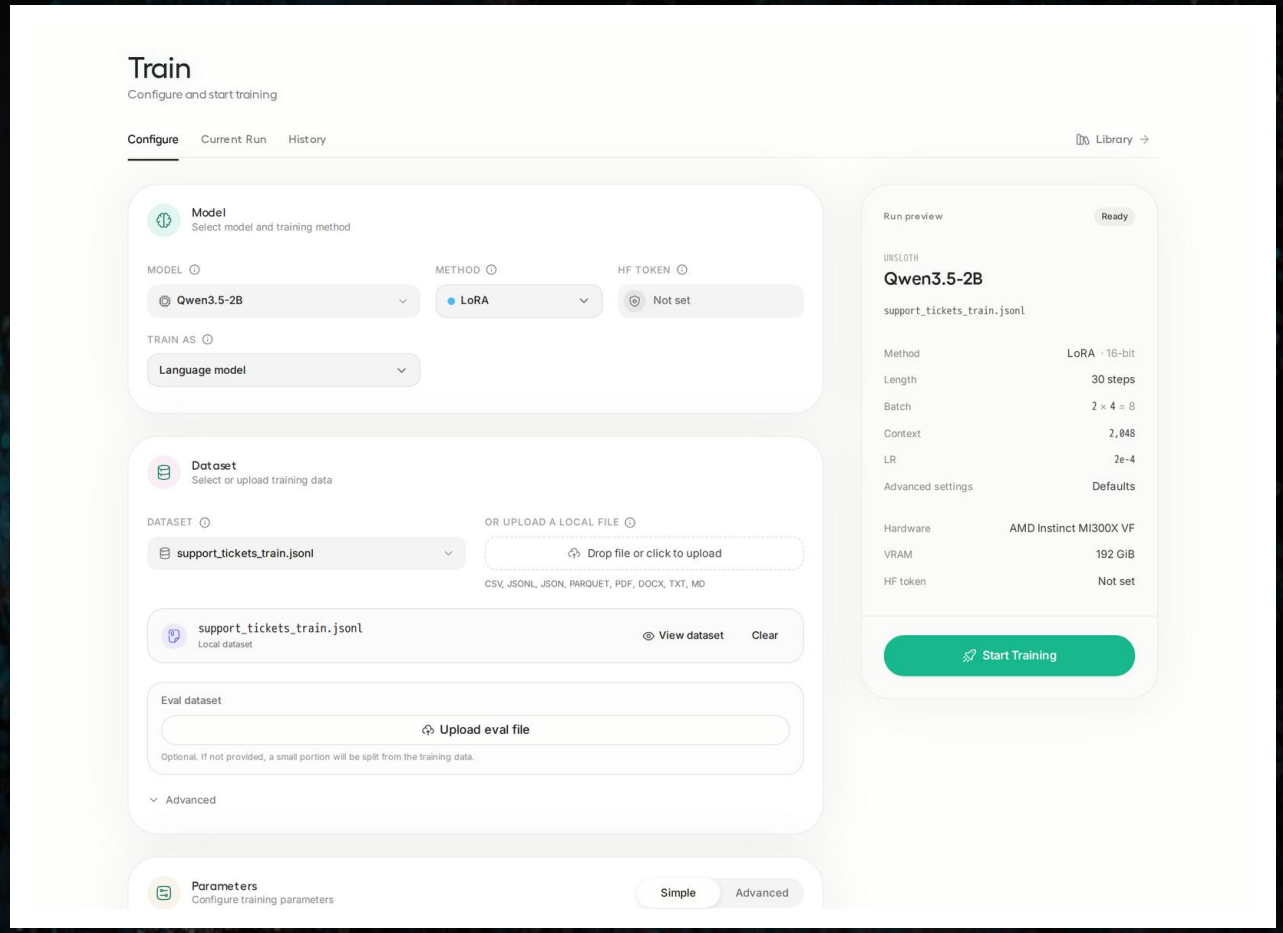
support_tickets_train.jsonl. View dataset should say ChatML

Set 30 steps, then Start Training

Watch the loss chart, and the GPU in `rocm-smi`

Compare in Chat

Thinking off, then paste a held-out message. Base and fine-tuned answer side by side



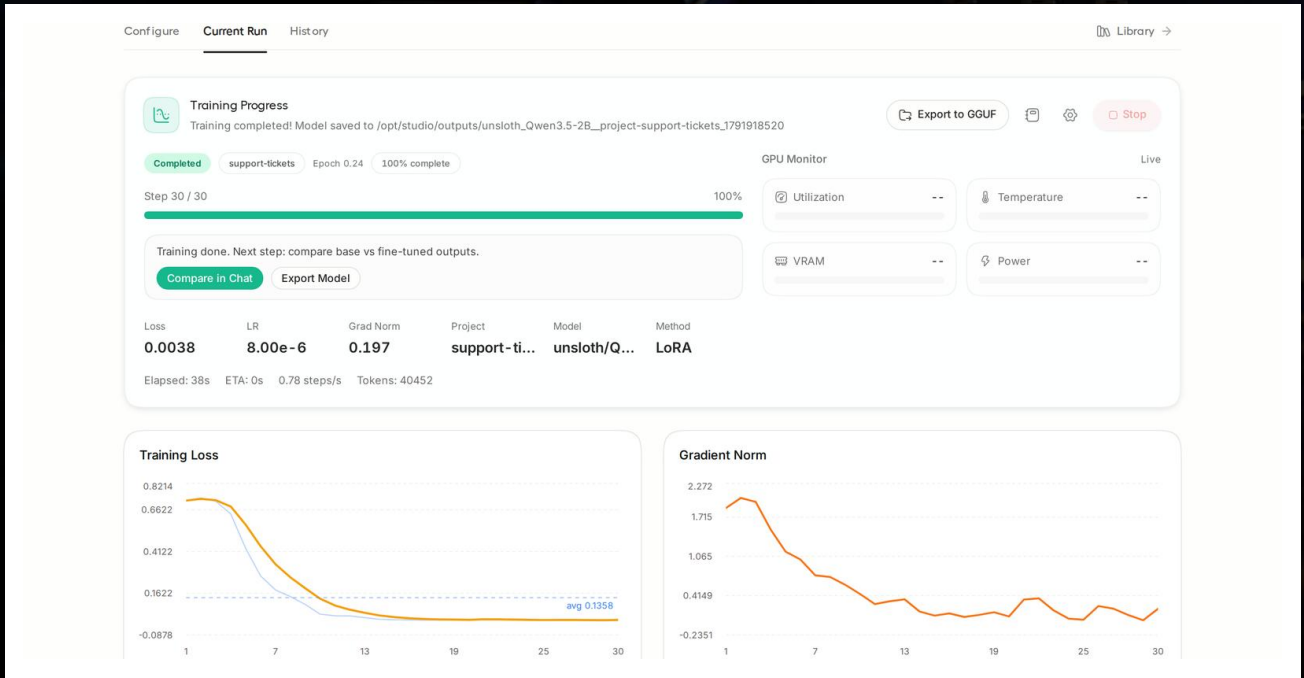
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Fine-tuning complete

Thirty SFT steps on a Radeon GPU, and the model speaks your schema.

DEMO: 30 STEPS



Loss and gradient norm per step, straight from Studio

{"order_id": "ORD-10968", "customer": "Elena", "phone": "(815) 781-9664", "product": "rice cooker", "category": "wrong_item", "purchase_date": "2026-01-24", "amount": 867.25}

25.9 tok/s

Fine-tuned answer to a held-out message, in exactly the schema

What to look for

- Loss drops in the first 15 steps, then flattens
- Held-out tickets exactly right on a Ryzen AI Max: 63% with a schema prompt, 97% fine-tuned

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Troubleshooting

SYMPTOM	FIX
First chat response times out	Cold model load. Retry, the warm model answers fast.
Dataset not read as chat	View dataset should say ChatML. Upload the .jsonl from data/ as is.
Out of memory during training	Smaller model, enable QLoRA 4-bit, lower batch size or sequence length.
Fine-tuned answers ramble	Turn Thinking off in Chat. Still rambling? Train more steps.
Training fails at step 0	Read the error in Current Run. A mismatched Triton install is a common cause.
Need a next experiment?	Your own JSONL with a messages column, or a bigger model like Qwen3.5-9B.

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Learn more

Unsloth	github.com/unslothai/unsloth	The framework itself
Unsloth docs	docs.unsloth.ai	Guides, notebooks, supported models
LoRA paper	arxiv.org/abs/2106.09685	Low-Rank Adaptation, Hu et al.
QLoRA paper	arxiv.org/abs/2305.14314	4-bit finetuning, Dettmers et al.
Qwen3.5-2B	huggingface.co/unsloth/Qwen3.5-2B	The model we fine-tune
Workshop notebook	notebooks.amd.com/ind-dev	Password: RadeonUnslothTuning

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